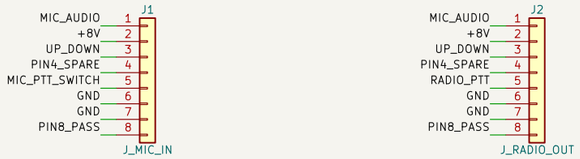
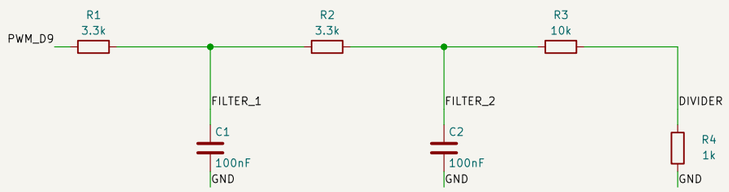
IC-229H CTCSS INTERFACE – 5V PRO MICRO / 151.4 Hz

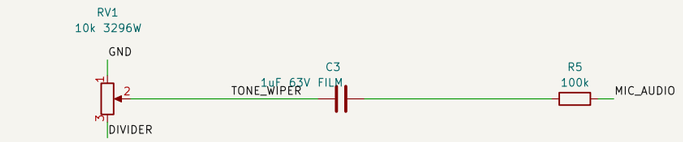
Schematic capture complete. CNC / assembly / radio bench acceptance still required; see implementation notes.



8-PIN PASS-THROUGH: pin 5 intercepted; pins 6/7 share GND.
Pin 4 and pin 8 designations require radio/cable verification.

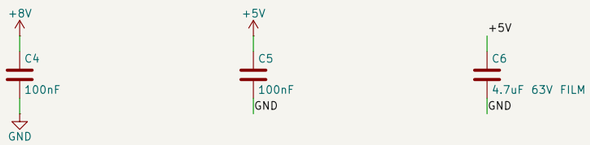
RC FILTER: 151.4 Hz tone; nominal PWM 10 kHz. DIVIDER is loaded by final level control.



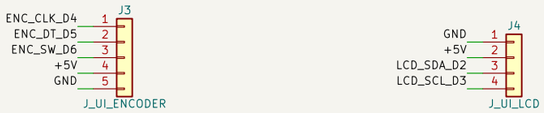


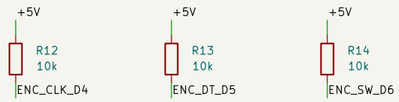
TONE LEVEL
3296W-1-103LF
Clockwise increases.
Start fully CCW.

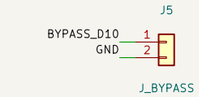
SUPPLY: +8V directly to U1 RAW; U1 VCC supplies +5V.
C4 RAW decoupling; C5 +5V decoupling; C6 4.7uF TDK film, 15mm pitch.
C3 1uF KEMET film, 5mm pitch. Both film capacitors nonpolar, 63V.



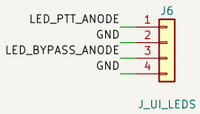
OFF-BOARD UI: pin order follows specification. Switches close to GND.
10k external pull-ups provided; verify onboard encoder pull-ups and LCD I2C pull-ups.

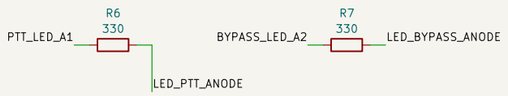


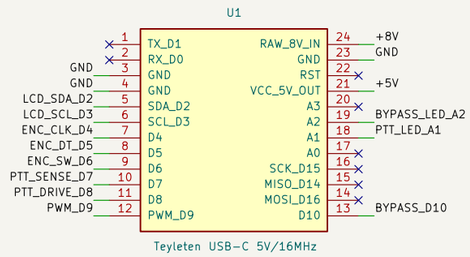










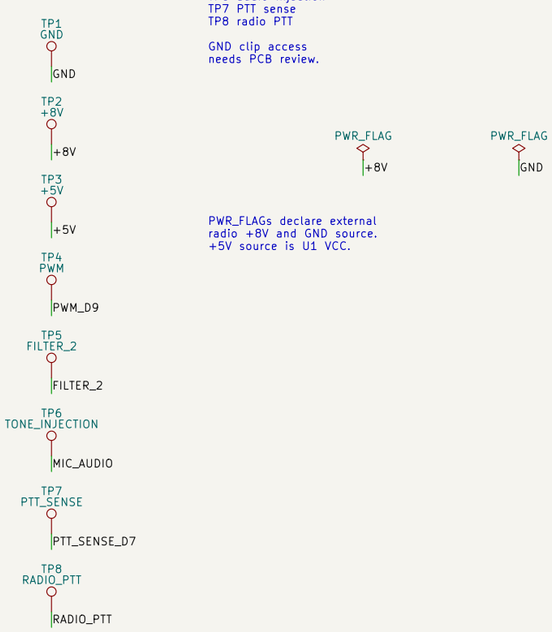


IMPLEMENTATION / BRING-UP
Bypass disables tone; PTT depends on powered, functioning firmware (firmware out of scope).
CNC: 1.5mm single-sided blank; bottom copper only; no planes/vias/jumpers. Physical marking may be external.
Radio wiring/current, LCD/backlight load, onboard I2C pull-ups, deviation and PWM noise require bench checks.
Use current-limited supply, inspect shorts and socket orientation, then insert module with power off.

PTT SENSE / DRIVER: default PTT released during reset.
Q1 onsemi 2N3904TAR, factory-formed 2.54mm leads; E=1, B=2, C=3.
Initial sink target <=5mA; verify actual radio voltage/current and saturation.

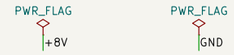


MODULE: user-approved nominal Biacco42 pad map, 2.54mm pitch / 15.24mm rows.
CNC derivative retains all 24 centers; enlarged B.Cu pads. Mount in female sockets.
Reserve USB-C / clone / cable access; verify physical fit before soldering.
Do not connect radio power and USB simultaneously until module power paths are verified.



TEST POINTS
TP1 GND
TP2 RAW / +8V
TP3 VCC / +5V
TP4 PWM
TP5 filtered tone
TP6 audio injection
TP7 PTT sense
TP8 radio PTT

GND clip access
needs PCB review.



PWR_FLAGs declare external
radio +8V and GND source.
+5V source is U1 VCC.

Sheet: /		
File: icom-ctcss-board.kicad_sch		
Title:		
Size: A3	Date:	**Rev:**
KiCad E.D.A. 10.0.0		Id: 1/1